

Method 3: Connect any one conductor to sheath, measure capacitance between remaining two conductors

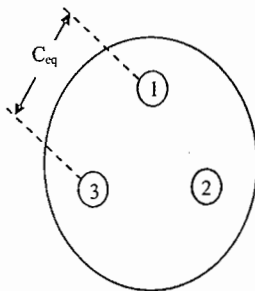
$$C_3 = \frac{C_s + 3C_c}{2} = \frac{C_n}{2}$$

$$C_n = 2C_3$$

Example 2.7.2:

A 40 km long cable has a capacitance of 0.15 $\mu\text{F}/\text{km}$ between any two conductors when the other conductor is connected to sheath. The supply voltage is 33kv, 50Hz, 3- ϕ . Determine the charging current I_{ph} .

Sol:



$l = 40 \text{ km}$

$$C_{eq} = 0.15 \mu\text{F}/\text{km}$$

$$C/ph = 2 \times C_{eq}$$

$$= 2 \times 0.15 = 0.3 \mu\text{F}/\text{km}$$

$$I_c / ph = \frac{V_{ph}}{X_{eq} / ph} = \frac{33\text{K} / \sqrt{3}}{\left(\frac{1}{\omega C} \right) / ph}$$

$$= \frac{33\text{K}}{\sqrt{3}} 2\pi f C / ph$$

$$= \frac{33}{\sqrt{3}} \times 10^3 \times 2\pi \times 50 \times 0.3 \mu\text{F}/\text{km}$$

$$= 1.79 \text{ A}/\text{km}$$

$$\therefore I_c / ph = 1.79 \times 40 = 71.8 \text{ A}$$

2.8 Over Head Insulators

The insulators for over head lines provide insulation to the power conductor from the ground. The insulators are connected to the cross arm of the supporting structure and power conductor passes through the clamp of the insulator. The insulator is to avoid leakage of current through the supports to the earth. Thus the insulators play important role in the successful operation of over head lines.

Every insulator must have following properties:

1. Mechanically very strong in order to withstand the load due to the weight of the conductor wind load etc.
2. High insulating resistance in order to avoid leakage of currents to earth.
3. High relative permittivity in order to give high dielectric strength.
4. High ratio of *puncture* strength to flash -over voltage.

5. The material should not be porous and there should be no effect of change in Temperature.
6. It must be free from internal impurities, cracks.
7. It should be impurities to fluids and gases of the atmosphere.

Materials used:

1. Porcelain
2. Glass
3. Steatite
4. Synthetic resin

Porcelain:

1. Now a days it is used extensively for the manufacture of insulators.
2. It is produced by firing a mixture of Quartz, Kaolin and Feldspar at a high temperature.
3. It is mechanically stronger than glass.



4. Its surface is not affected by dirt deposits and it withstands temperature changes.
5. It has dielectric strength of about 60 kV/cm and a compressive strength of 7000 kg/cm²
6. It is used upto 33 kV.

Glass:

1. Glass is cheaper than porcelain in simpler shapes.
2. It gives high resistivity.
3. Dielectric strength of about 140 kV/cm.
4. It has higher compressive strength than porcelain.
5. Its coefficient of thermal expansion is less.
6. It is used upto 25 kV under ordinary atmospheric conditions and upto 50 kV in dry atmosphere.

Steatite:

1. Steatite is a naturally occurring magnesium silicate usually found combined with oxides in varying proportions.
2. It has a much higher tensile and bending stress than porcelain.
3. These insulators are used at tension towers or when transmission line takes sharp turn.

Classification of insulators:

1. Pin insulators
2. Shackle insulators
3. Suspension insulators
4. Strain insulators

Pin insulators:

1. It can be employed for operating voltages upto about 25 kV satisfactorily.
2. An adequate length of leakage path is obtained by providing the insulator with two or three petticoats or rain sheds.
3. Multipart pin type insulators are used upto 33 kV and beyond which their weight is more and uneconomical.
4. The cost of pin insulators increases much more rapidly than the voltage.

$$\text{Cost} \propto v^x \quad \text{where } x > 2$$

- Pin type insulators are not suitable for voltages higher than 33 kV.

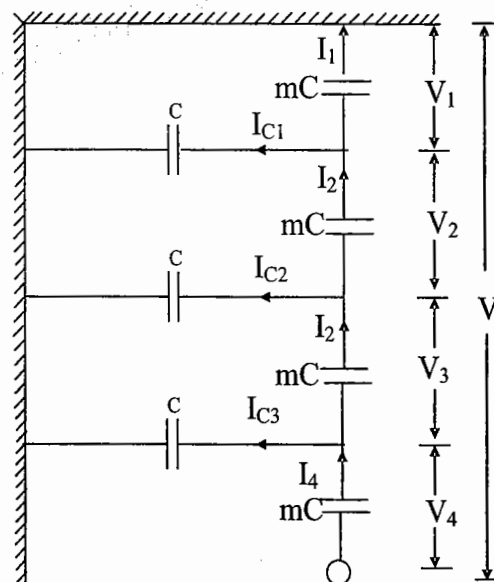
Suspension insulators:

1. Suspension type insulators are used for higher voltages (beyond 33kV).
2. These insulators consists of one or more insulators units flexibly connected together and adopted to be hung for the cross-arm of the supporting structure and carry a power conductor at its lowest extremity is known as string insulators.
3. Each insulator is designed for 11 kV and hence for any operating voltage a string of insulators can be used. For example for 132 kV transmissions, the number of insulators required is 12
4. In case of failure of one of the units in the string, only that particulars unit needs replacement rather than the whole string.
5. Low mechanical stress.
6. The operating voltage of the existing transmission can be increased by adding suitable number of discs in the string instead of replacing all the insulators as it is necessary in case of pin type insulators.
7. It is used horizontal plane rather than vertical plane.

Voltage distribution across a string of suspension insulators:

A string insulator may be treated as a combination of capacitances connected as shown in fig.

Let V = operating voltage per phase





V_1, V_2, V_3, V_4 = voltage across various units
 $\therefore V = V_1 + V_2 + V_3 + V_4$ (1)

$$m = \frac{\text{Mutual capacity}}{\text{Capacitance to ground}}$$

In the fig.,

$$\begin{aligned} I_2 &= I_1 + I_{c1} = V_1 m \omega C + V_1 \omega C \\ &= V_1 \omega C (m + 1) \end{aligned}$$

$$\therefore V_2 = \frac{m+1}{m} V_1 \text{ (2)}$$

$$I_3 = I_2 + I_{c2}$$

$$\Rightarrow V_3 m \omega C = V_2 m \omega C + (V_1 + V_2) \omega C$$

$$\Rightarrow V_3 m = V_2 (m + 1) + V_1$$

From Equation (2),

$$\Rightarrow V_3 m = \frac{(m+1)^2 V_1}{m} + V_1 \frac{m^2 + 3m + 1}{m}$$

$$\therefore V_3 = \frac{m^2 + 3m + 1}{m} V_1 \text{ (3)}$$

Similarly V_4 can be expressed as,

$$I_4 = I_3 + I_{c3}$$

$$\Rightarrow V_4 m \omega C = V_3 m \omega C + (V_1 + V_2 + V_3) \omega C$$

$$\Rightarrow V_4 m \omega C = \frac{m^2 + 3m + 1}{m} V_1 + \frac{V_1 + V_1(m+1)}{m} +$$

$$\frac{m^2 + 3m + 1}{m} V_1$$

$$\therefore V_4 = \frac{m^3 + 6m^2 + 5m + 1}{m} V_1$$

From the above four equations, values of potential across each unit can be determined.

- The voltage drop across the unit nearest the cross arm is minimum and it goes on increasing as we go towards the power conductor
- The voltage drop across the unit nearest the power conductor is maximum.
- Mutual capacitance of each disc is same.
- The current through the top most unit is minimum. Current goes down the unit towards

power conductor goes on increasing being the maximum.

$$\begin{aligned} \text{String efficiency} &= \frac{\text{voltage across the string}}{n \times \text{voltage across the unit}} \\ &\quad \text{near the power conductor} \\ &= \frac{\text{sov for the string}}{n \times \text{sov of one disc}} \end{aligned}$$

Where 'n' is the number of insulators in the string. sov is spark over voltage.

- String efficiency always less than one or unity.
- The voltage across various units will tend to be equal in case the value of 'm' is large.
- In the case of high voltage lines since the clearance between the conductor and the tower structure should be more to avoid flash over under normal operating condition.
- The mutual capacity being fixed the ground capacitance goes on depending with larger clearance and hence the ratio of the two capacitances goes on increasing.
- The unit nearest the cross-arm should have the minimum capacitance maximum reactance and as we go towards the power conductor the capacitance should be increase by using grading equalizing the potential all units.

Disc type insulators are used in different voltages:

Voltage kV	66 kV	132 kV	220 kV	400 kV
No. of discs	4 - 5	9 - 10	15 - 16	22 - 23

Strain insulators:

These insulators are special mechanically strong disc insulators used to take the tension of the conductors at line terminals and at point where the line is dead end. For example road crossing, junctions of overhead lines with cables, river crossing, at angle towers where there is change in line.

Shackle insulators:

These are used for L.T lines, these are used in section poles, end poles, on sharp turn and for service connection. These can be used either in a horizontal or in a vertical position.



String efficiency can be improved:

1. By using longer cross – arms.
2. By grading the insulators.
3. By using a guard ring.

Grading is served two purposes:

1. Equalization of voltage drop across the units.
2. When used with arcing horn it protect the insulator string from flash over whenever an over voltage (under normal or abnormal condition) appear between the tower structure and power conductor.

Causes of Failure of Insulators:

1. Deterioration by cracking of the porcelain.
2. Porosity.
3. Puncture of weak porcelain.
4. Shattering by power arcs.
5. Flash over caused by dust deposits.
6. Failure from mechanical stresses.
7. Short circuits by birds and similar objects.

Example 2.8.1:

Match List-I (Insulation Type) with List-II (Purpose or Configuration) and select the correct answer using the codes given below the lists: (IAS-01)

List-I (Insulation Type)

- A. Pin type
- B. Suspension type
- C. Strain type
- D. Shackle type

List-II (Purpose or Configuration)

1. Low voltage distribution lines
2. String of insulators in horizontal position
3. String of insulator in vertical position
4. For voltage up to 33 kv

Codes:

	A	B	C	D
(a)	4	2	3	1
(b)	1	3	2	4
(c)	4	3	2	1
(d)	1	2	3	4

Sol: (c)

Example 2.8.2:

If the voltage across the string of a string insulator assembly is 38 kV, number of insulation discs are 4 and voltage across the lowermost disc is 12 kV, the string efficiency is

Sol: Total voltage $V = 38$ kV

Number of discs, $n = 4$

Last disc voltage, $V_4 = 12$ kV

$$\eta_{\text{string}} = \frac{V}{4 \times V_4} = \frac{38}{4 \times 12}$$

$$= 79.17\%$$

Example 2.8.3:

In a three unit insulator string, voltage across the lowest unit is 17.5 kV and string efficiency is 84.28%. the total voltage across the string will be equal to _____

Sol: $n = 3$, $V_3 = 17.5$ kV

$$\eta_{\text{string}} = 84.28\%$$

$$\text{So, } \eta_{\text{string}} = \frac{V}{3 \times V_3}$$

$$0.8428 = \frac{V}{3 \times 17.5}$$

Total voltage, $V = 44.25$ kV

Example 2.8.4:

A string insulator has 4 units. The voltage across the bottom – most unit is 33.33% of the total voltage, its string efficiency is _____

Sol: $n = 4$

$$V_4 = 33.33\% \text{ of } V$$

$$= 0.333V$$

$$\eta_{\text{string}} = \frac{V}{4 \times V_4}$$

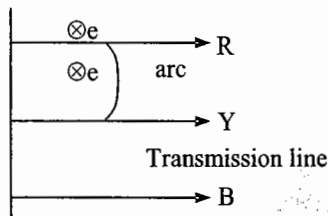
$$= \frac{V}{4 \times 0.333V}$$

$$= 75 \%$$

2.9 Concept of Corona

Corona: The ionization of Insulating material (air) surrounding the surface of the conductor of a transmission line (or) the disruption of the dielectric strength of air near the conductor of a transmission line.

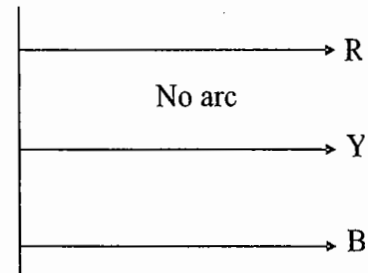
Description:



On economical basis, most of the transmission system is overhead system. There are certain free electrons which are available in the space in which some of the electrons are very near to surface of the conductor and some of them are away from the surface conductor. The electrons near the power conductor will move at a faster rate than that of electron away from the surface of the conductor, because the electric field intensity is maximum at the surface of the conductor. The electric field intensity will be proportional to the operating voltage of the system.

Whenever the operating voltage is more than 270 kV, the field intensity will also increase so that the mobility of the electron will increase. The high mobility electron will collide with the other air molecules, which will dislodge the electrons from those molecules. The process is continuous and at certain instant the electron avalanche will take place i.e., an arc will strike between the two lines. The arc is an ionized body, which will be treated as line to line fault. The reliability of supply is reduced.

In order to eliminate the arc, the distance between the two lines is increased in a non-linear manner as the operating voltage increases, so that the field intensity will be restricted to be maximum at the surface of the conductor and almost zero between the two conductors.

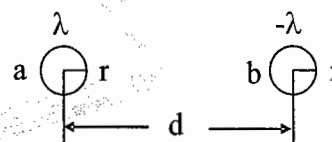


However the conductors which are employed for transmission line are stranded conductors having rough surface. Due to rough surface certain impurities are to be deposited. So a self sustained field emission will be initiated around the surface of the conductor at certain points. The voltage at which the self sustained discharge will be initiated is called the critical disruptive voltage.

The corona initiation can be identified as

- Hissing noise.
- Releasing of ozone gas.
- Occurrence of beds and Tufts.

Critical disruptive voltage of a 1-phase 2-wire transmission line:



$$\text{Critical disruptive voltage } V_d = g_0 r \ln \left(\frac{d}{r} \right)$$

r = radius of conductor in cm

d = distance of separation in m.

g_0 = Dielectric strength of air

= 30 kV/cm (peak)

= 21.1 kV/cm (rms) at NTP

However the existing temperature and pressure at the surface of the conductor are different from NTP. So at any other temperature and pressure.

$$V_d = g' r \ln \left(\frac{d}{r} \right) \text{ kV/rms}$$

g' = Dielectric strength at any temperature and pressure.

$$g' = g_0 \delta$$

δ = air density factor

$$\delta = \frac{3.92 \times h}{273 + t}$$

δ = air density factor

h = Atmospheric pressure in cm of Hg.

t = temperature in $^{\circ}\text{C}$.

If the surface of the conductor is irregular then surface irregularity factor (m) will be introduced.

$$V_d = 21.1 m \delta r \ln \left(\frac{d}{r} \right) \text{ kV/rms}$$

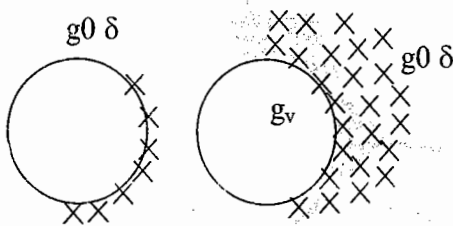
$m = 1.0$ for smooth conductor

$m = 0.93 - 0.98$ for rough (or) stranded conductor

$m = 0.83 - 0.87$ for seven stranded cable

$m = 0.9$ for more than seven stranded cable

Visual critical Disruptive voltage: when the corona is initiated it will not be visible. In order to visualize the corona, further ionization needs to be done at the surface of the conductor.



$$g_v = g_0 \left(1 + \frac{0.3}{\sqrt{r \delta}} \right)$$

The visual critical disruptive voltage

$$V_v = 21.1 m_v r \delta \left(1 + \frac{0.3}{\sqrt{r \delta}} \right) \ln \left(\frac{d}{r} \right) \text{ kV/rms}$$

m_v = surface irregularity factor

$m_v = 1.0$ for smooth surface of the conductor

$m_v = 0.93 - 0.98$ for rough conductor (or) stranded conductor

The visual corona will be observed as white bluish colour.

Example 2.9.1:

Under NTP conditions the dielectric strength of air is 30 kV/cm (peak) what is the dielectric strength of air at a pressure of 72 cm of Hg and at a temperature of 30°C .

Sol: at any t , h dielectric strength of air is $g = g_0 \delta$

$$\delta = \frac{3.92h}{273 + t}$$

At $h = 72$ cm at Hg, $t = 30^{\circ}\text{C}$

$$\delta = \frac{3.92 \times 72}{273 + 30} = 0.9434$$

$$g = 30 \times 0.9434 \text{ kV/cm (peak)} \\ = 28.302 \text{ kV/cm (peak)}$$

Critical Disruptive voltage for 3-phase system:

$$V_d = 21.1 m \delta r \ln \left(\frac{\text{GMD}}{r} \right) \text{ kV/rms/phase}$$

GMD = Mutual distance

Local corona: Ionization of air at certain points along the power conductor when the conductor is deposited with ice. The 'm' value for local corona is 0.72.

Decided Corona: The visual corona at certain points when the conductor is deposited with ice. The 'm' value for decided corona will be 0.8.

Corona loss: When the corona initiated on the transmission lines, certain real power loss will take place. The amount of real power loss is called as corona loss.

$$P = 241 \times 10^{-5} \left(\frac{f + 25}{\delta} \right) \sqrt{\frac{r}{d}} (V_{ph} - V_d)^2 \text{ kW /ph /km}$$

f = supply frequency in Hz

δ = air density factor

r = radius of conductor in cm.

d = distance of separation in m.

V_{ph} = operating voltage / phase/ rms.

V_d = critical disruptive voltage rms/phase

Note: When the operating voltage is more than critical disruptive voltage, then only the corona will be initiated. Other wise there is no corona.

Eg: $f = 50$ Hz, $r = 3$ cm, $d = 8$ m,

$\delta = 0.97$, $V_d = 105$ kV / phase,

$V_p = 100$ kV/phase.

The corona loss will zero



Example 2.9.2

At 50Hz operating frequency a HVAC transmission has a corona loss of 1.2 kW/ph/km. What is the corona loss of the same line operating at 60Hz.

Sol: $P_{\text{loss}} \propto (f + 25)$

At $f_1 = 50\text{Hz}$, $P_{\text{loss}} = 1.2 \text{ kW/ph/km}$

At $f_2 = 60 \text{ Hz}$,

$$P_{\text{loss } 2} = P_{\text{loss } 1} \times \frac{f_2 + 25}{f_1 + 25}$$

$$= 1.2 \times \frac{60 + 25}{50 + 25} = 1.36 \text{ kW/ph/km}$$

Factors Influencing corona loss:

1. Electrical factors:

- Supply frequency increases the corona loss will increase, because $P \propto (f+25)$.
- The corona loss of Ac Transmission line is more than DC transmission line for the same operating voltage.
- The corona loss at positive polarity conductor at a DC transmission line is more than negative polarity conductor. At the positive polarity conductor the electron will move at a faster rate due to positive electric field Intensity and negative polarity electron. So the critical disruptive voltage is less and corona loss will be high.
- when the sinusoidal wave form is having distortion (i/e) consists of harmonics, the corona loss will increase.
- lower the height at the power conductor, higher will be the corona loss. Due to lower height, the capacitance of the conductor will increase so that the operating voltage will increases. The increased operating voltage will provide high electric field intensity so that the critical disruptive voltage is less which in turn increases the corona loss.
- Conductor configuration:** when the transmission lines are arranged in equilateral form, the capacitance / phase will be same so the operating voltage will be same, which in than the critical disruptive voltage will be same for the three conductors. So the corona

loss will be same. In case of unsymmetrical conductor configuration, the capacitance of the middle conductor is high, so the critical disruptive voltage is less and the corona loss is high.

2. Atmospheric factors:

- Temperature and pressure: Both are related to air density factor.

$$\delta = \frac{3.92 \times h}{273 + t}$$

$$P \propto 1/\delta \text{ (or) } P \propto (V_p - V_d)^2 \text{ (or) } V_d \propto \delta$$

As 'δ' increases the corona loss will decreases. As δ increases V_d will increases so $(V_p - V_d)^2$ will reduce which will reduce corona loss.

In hill area the corona loss is high when compared to sea level area.

$$\delta = \frac{3.92 \times h}{273 + t}$$

In hill area the falling of pressure is very high when compared to falling of temperature. So 'δ' will be less in hill area and corona loss will be high.

- Deposition of Dust, ice, snow and for:** These are all impurities to the surface of the power conductor, which will reduce the dielectric strength of air so that the critical disruptive voltage will reduce and the corona loss will increase.

3. Factors related to the size of the conductor:

- use larger diameter of the conductor:

$$P \propto \sqrt{r} \dots (1)$$

$$P \propto (V_{ph} - V_d)^2 \dots (2)$$

$$V_d \propto r \dots (3)$$

As the diameter increases, there is a possibility to increase the corona as per equation

- However in equation
 - it will be reducing, because V_d will increase a per equation
 - The fall in $(V_{ph} - V_d)^2$ is more predominant than that of raise in $\sqrt{r}$ term.
- Use more than conductor per phase: By using more than are sub-conductor/phase the self



distance will increase. So the critical disruptive voltage will increase. The corona loss is reduced.

- (c) **Profile of the conductor:** By using cylindrical shape of conductor the field is more uniformly distributed throughout the cross section at the conductor. So the V_d will increase and the corona loss will be reduced.

Disadvantages of Corona:

01. There is certain real power loss apart from ohmic loss.
02. When the corona is initiated on transmission lines, the overall diameter of the conductor will increase so that the capacitance will increase. The net charging flux which will link with the neighboring communication line is not zero. So there will be communication interference. Apart from the charging flux, the zero sequence components during ground fault will also cause interference.

Advantages: Corona will act as a safety valve against direct lightning stroke, by dissipating the peak magnitude of lightning stroke in the form of corona loss.

Methods of reducing corona loss:

- (1) use larger diameter of the conductor
- (2) use hollow conductor
- (3) use bundle conductor

Corona loss will be high on a stranded conductor and it will be less on a hollow conductor. However, the corona will exist beyond 275 kV transmission system only. So bundle conductors will be used in which each sub conductor is a stranded conductor.

Example 2.9.3:

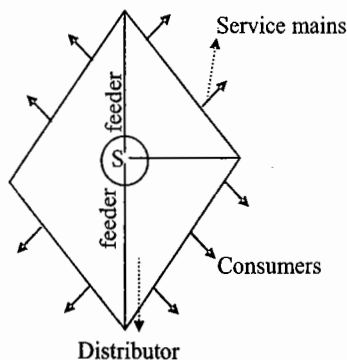
Which of the following equipment placed at the terminal of HV bushing of a HV power transformer to reduce corona loss,

- (a) Bundle conductor (b) hollow conductor
(c) stranded conductor (d) corona ring

Ans: (d)

2.10 Distribution System

The part of power system by which electric power is distributed among various consumers for their local use is known as distribution system. A low tension distribution system is shown in fig. below, which comprises the feeders, distributors and service mains.



1. **Feeder:** A line or conductor which connects the major station to the distributor is known as **feeder**.

2. **Distributor:** A line or conductor to which various consumers are connected through service mains is known as **distributor**.
3. **Service Main :** A line (conductor or cable) which connects the consumer to the distributor is known as **Service main**.

Classification of distribution system:

1. **As per the type of current**
 - i) D.C. distribution
 - ii) A.C. distribution
2. **According to construction**
 - i) Overhead distribution system
 - ii) Underground distribution system
3. **According to the service**
 - i) General lighting and power
 - ii) Industrial power
 - iii) Railway
 - iv) Street lighting etc



4. According to the number of wires

- i) Two wire
- ii) Three wire
- iii) Four wire

5. According to scheme of connections

- i) Radial system
- ii) Ring system
- iii) Interconnected system

DC Distribution : The electric power is almost exclusively generated, transmitted and distributed as a.c. but for certain applications (e.g for electro-chemical works, for the operation of variable speed machinery d.c. motors etc.) is absolutely necessary. For this purposes, a.c is converted into d.c. at the sub-station and is then distributed by i) 2 -wire system ii) 3 - wire system.

AC Distribution: The electric power (or energy) is invariably generated, transmitted and distributed in the form of alternating current. The main reason of adopting a.c. system for generation, transmission and distribution of electric power is that the alternating voltage can conveniently be changed to any desired value with the help of a transformer.

- i. **Primary distribution:** The system in which electric power is conveyed at 11kV or 6.6 kV or 3.3 kV to different sub-stations for distribution or to big consumers (e.g industries, factories etc) is called primary distribution system.
- ii. **Secondary distribution system:** The system in which electric power is distributed at 400/230 V to various consumers (e.g residential consumers) is called low voltage or secondary distribution system.

Connection schemes of distribution system :

Radial system: In radial system, separate feeders radiate from a single substation and feed the distributors at one end only.

Advantages:

- i) Its initial cost is minimum.
- ii) It is the simplest distribution system.

Disadvantages:

- 1. The end of the distribution nearest to the feeding end would be heavily loaded.
- 2. The consumers are dependent on single feeder and distribution. Therefore when fault occurs on the feeder or distributor, the supply is cut off to all the consumers who are on the side of the fault away from the sub-station .
- 3. The consumer at the farthest end of the distributor would be subjected to serious voltage fluctuations with the variation in load.

Ring main system: In this system each consumer is supplied via two feeders. The arrangement is similar to two feeders in parallel on different routes.

Advantages:

- 1. Less copper is required as each part of the ring carries less current than in radial system.
- 2. Less voltage fluctuations.
- 3. It is more reliable. In the event of fault on any one section, the continuity of supply to all consumers can be maintained by isolating the faulty section.

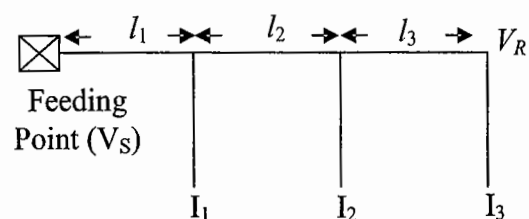
Inter connected system: In this system, the feeder ring is energized by two or more than two generating stations or substations.

Advantages:

- 1. It increases the reliability of supply.
- 2. During overload hours, the area fed from one generating station can be fed from other generating station. Thus it reduces the reserved plant capacity, improves the service reliability and increases the load factor and efficiency of the system.

Voltage drop in distributors:

A) Load at unity Power Factors :



If l_1, l_2, l_3 are length in meters.

I_1, I_2, I_3 are loads on the line in Amps.

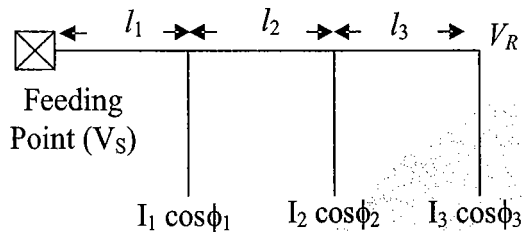


'r' is loop resistance of distributor in ohm per meter.

$$\text{Total voltage drop} = [(I_1 + I_2 + I_3) \cdot l_1 \cdot r] + [(I_2 + I_3) \cdot l_2 \cdot r] + [(I_3) \cdot l_3 \cdot r] \text{ volts.}$$

$$\text{Power-loss} = [(I_1 + I_2 + I_3)^2 \cdot l_1 \cdot r] + [(I_2 + I_3)^2 \cdot l_2 \cdot r] + [(I_3)^2 \cdot l_3 \cdot r] \text{ watts.}$$

B) Load at Different Power Factors:



If l_1, l_2, l_3 are length in meters.

I_1, I_2, I_3 are loads on the line in Amps.

'r' is loop resistance of distributor in ohm per meter

'x' is reactance of distributor in ohm per meter

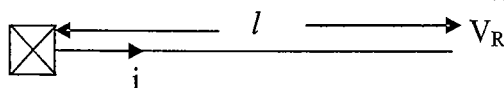
Total voltage drop

$$= [I_1(r \cos \phi_1 + x \sin \phi_1)l_1] + [I_2(r \cos \phi_2 + x \sin \phi_2)(l_1 + l_2)] + [I_3(r \cos \phi_3 + x \sin \phi_3)(l_1 + l_2 + l_3)] \text{ Volts}$$

Powerless

$$= [I_1^2(r \cos \phi_1 + x \sin \phi_1)l_1] + [I_2^2(r \cos \phi_2 + x \sin \phi_2)(l_1 + l_2)] + [I_3^2(r \cos \phi_3 + x \sin \phi_3)(l_1 + l_2 + l_3)] \text{ Watts}$$

C) Uniformly loaded distributed fed at one end:



Feeding Point (V_s)

l = total length of conductor ;

i = current tapped off per meter length ;

r = total resistance of distributor in ohm per meter;

x = reactance of distributor in ohm per meter

Where $il = I$ = Total current in amp ;

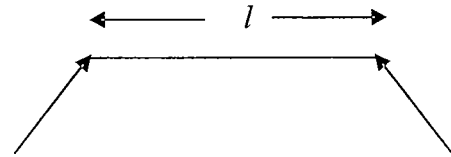
$lr = R$ = total resistance in ohm

$lx = X$ = total reactance in ohm;

$\therefore$ Total voltage drop

$$= (I/2)[R \cos \phi + X \sin \phi] \text{ volts}$$

D) Uniformly loaded distributed fed at both ends



l = total length of conductor ;

r = resistance of distributor in ohm per meter

x = reactance of distributor in ohm per meter

where $lr = R$ = total resistance in ohm $lx = X$

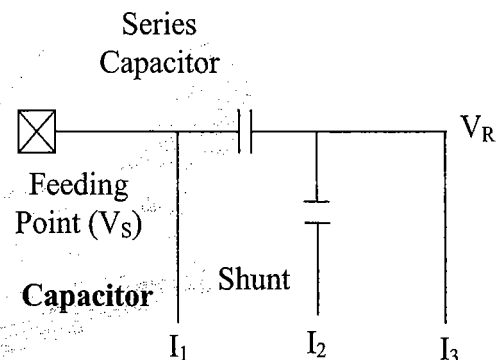
= total reactance in ohm;

The voltage drop upto middle

$$= (I/8) [R \cos \phi + X \sin \phi] \text{ volts}$$

Importance of capacitor in distribution system:

1. Voltage improvement
2. Loss reduction
3. Power factor improvement



- Series capacitor influence is down streamside and where as in case of shunt capacitor influence is upstream side.
- Series capacitor (AVB) is mainly used in voltage improvement and Shunt capacitor is mainly used in loss reduction and power factor improvement.
- Automatic Voltage Boosters are used for voltage improvement and these are low cost, marginally loss reductions.